

WHEAT

A Generalized Vernalization Response Function for Winter Wheat

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ABSTRACT

Vernalization is a process required for certain plant species to enter the reproductive stage through an exposure to low, nonfreezing temperatures. These plant species include some fall-planted cereals, among them winter wheat (*Triticum aestivum* L.). A three-stage linear function is currently used in wheat simulation models to describe the developmental response to the duration of the vernalization treatment, expressed as effective vernalization days (VD). This function lacks generality because the value of its coefficients varies with genotype. The objective of this study was to develop a generalized nonlinear vernalization response function for winter wheat. The nonlinear vernalization function developed in this study has coefficients with biological meaning. Data of final leaf number at different VD treatments in 12 winter wheat cultivars from 19 trials, which are from published research and from a growth chamber experiment conducted as part of this study, were used as independent data for evaluating the nonlinear vernalization function. These data sets represent a wide range of winter wheat cultivars developed in different parts of the world. The generalized nonlinear vernalization function described the developmental response to VD better (RMSE = 0.032) than the three-stage linear functions (RMSE = 0.060 for cultivar Karl 92 and RMSE = 0.129 for cultivar Arapahoe). It is concluded that the vernalization response of winter wheat can be described by a general vernalization function. This conclusion implies that a reduction in the input data requirements is possible for winter wheat simulation models.

WINTER WHEAT requires exposure to low, nonfreezing temperatures before flowering can occur (Paulsen, 1987). The exposure to low temperatures, either in natural winter or in artificial cold treatment, that causes induction of flowering is called vernalization (Flood and Halloran, 1986). Vernalization response is common to fall-planted winter cereals and is a survival mechanism to tolerate low temperatures. The first description of a cold requirement for winter cereals was by G. Gassner in 1918 (see review by Purvis, 1961). The work by Purvis (1939, 1947, 1948) and Purvis and Gregory (1937, 1952) provided the hypotheses and the basis for what is currently known about the vernalization response. Since then, the effect of vernalizing temperatures on winter wheat has been widely studied, and the factors that influence vernalization response have been identified, i.e., temperature range over which the vernalization response takes place, duration of the vernalization period, genotype responses, and plant age (Chujo, 1966; Jedel et al., 1986; Wang et al., 1995; Rawson et al., 1998). Some as-

pects of the vernalization response, however, are still under debate, such as the expression of the vernalization response [should it be in units of thermal time to flowering or final leaf number (FLN)?] and how genetic variability should be characterized (Wang et al., 1995; Rawson et al., 1998).

Wheat plants respond to vernalization by decreasing their time to flowering (i.e., there is an increase in the development rate toward flowering), and there is no developmental response to vernalizing temperatures after flowering (Slafer and Rawson, 1994; Cao and Moss, 1997; Wang and Engel, 1998). The decrease in the time to flowering is caused by a decrease in the number of primordia that become leaves, i.e., a decrease in FLN (Hay and Kirby, 1991; Slafer and Rawson, 1994; Wang et al., 1995). The plant response to vernalization is given by the combination of two factors, the temperature during the vernalization period and the duration of the vernalization period. Vernalization has three cardinal (minimum, optimum, and maximum) temperatures (Yan and Hunt, 1999). In an extensive review, Porter and Gawith (1999) suggested that the cardinal temperatures for vernalization in wheat are -1.3 , 4.9 , and 15.7°C , respectively. The duration of the exposure to vernalizing temperatures is measured as VD. One VD is attained when the plant is exposed to the optimum temperature for vernalization for a period of 1 d (24 h). As temperature departs from the optimum, only a fraction of 1 VD is accumulated by the plant at a given calendar day (Hodges and Ritchie, 1991; Ritchie, 1991).

Because of its direct and indirect (through vernalization) effects on flowering time, temperature is one of the major environmental factors that affect the development rate in winter wheat, along with photoperiod. To account for the effect of VD on the development rate, several well-known crop simulation models use a response function for VD [vernalization function, $f(V)$], which varies from 0 to 1, as a modifier of the development rate (Weir et al., 1984; Ritchie, 1991; Wang and Engel, 1998).

Nonlinear response functions for temperature and photoperiod have been used in wheat simulation models (Angus et al., 1981; Shaykewich, 1995; Wang and Engel, 1998; Yan and Hunt, 1999). A literature search, however, yielded no indication of use of a nonlinear function for $f(V)$ in current models. Instead, $f(V)$ is typically modeled by a three-stage linear function (Weir et al., 1984; Reinink et al., 1986; Ritchie, 1991; Cao and Moss, 1997; Wang and Engel, 1998). The three-stage linear function has two coefficients: the minimum or base VD (VD_b),

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Abbreviations: FLN, final leaf number; MMF, Morgan–Mercer–Flodin (function); RMSE, root mean square error; VD, effective vernalization day(s); VD_b , base vernalization days; VD_{full} , full vernalization days.

defined as the VD below which no development occurs, and the maximum or full VD (VD_{full}), defined as the VD above which the development rate is maximum. The response function $[f(V)]$ is 0 when $VD \leq VD_b$ and then increases linearly to 1 when VD_{full} is achieved, and for any $VD \geq VD_{full}$, $f(V) = 1$.

The three-stage linear approach lacks generality because the end points, VD_b and VD_{full} , are genotype dependent (Weir et al., 1984; Hodges and Ritchie, 1991; Cao and Moss, 1997; Wang and Engel, 1998). This is a disadvantage because these two coefficients are unknown for many of the existing cultivars and for the new cultivars released every year, thus necessitating controlled experiments to quantify them, which are expensive and time and labor demanding. Another disadvantage of the three-stage linear approach is that it is composed of a combination of linear equations that introduces abrupt changes at the transition points of the response function. Wheat developmental response to VD is sigmoidal, which causes a significant departure from linearity (Chujo, 1966; Wang et al., 1995; Brooking, 1996; Rawson et al., 1998). Therefore, although attractive because it is simple to implement, the three-stage linear approach may not be the most appropriate response function for $f(V)$.

Our hypothesis was that most of the variability in the vernalization response among winter wheat cultivars can be represented by a generalized function. The objective of this research was to develop a generalized nonlinear vernalization response function for winter wheat. A generalized function has the advantage of being independent of cultivar, thus reducing the input data necessary in crop simulation models. A nonlinear function also may be more realistic from a biological point of view.

MATERIAL AND METHODS

Vernalization Response Function

Literature data on the response of wheat development (time to flowering or FLN) to VD show a consistent pattern (Chujo, 1966; Wang et al., 1995; Brooking, 1996; Rawson et al., 1998; Mahfoozi et al., 2001). When plants are exposed to less than about 8 to 10 VD, they do not differ from unvernallized plants. As the exposure to vernalizing temperatures progresses, plants tend to shorten their time to flowering or decrease their FLN in an increasing rate as VD increases up to about 15 to 20 VD. By continuing the exposure to vernalizing temperatures, the time to flowering or FLN decreases with VD in a linear fashion up to about 35 VD. After 35 VD, the response (decrease in time to flowering or FLN) slowly decreases as VD approaches 50 VD when plants are fully vernalized. After 50 VD, the response to VD saturates.

These results suggest a sigmoidal-shaped curve for describing the developmental response of wheat to VD. There are several functions with a sigmoidal shape. Among them, a flexible sigmoidal response function is the Morgan–Mercer–Flodin (MMF) function (Morgan et al., 1975):

$$Y = (ab + cX^n)/(b + X^n) \quad [1]$$

where Y is the dependent (or response) variable, X is the independent (or explanatory) variable, a is the intercept when $X = 0$, c is the asymptote as X approaches infinity, n is a shape coefficient, and b is interpreted as $b = (X_{0.5})^n$, with $X_{0.5}$ being the value of X when Y is half of the maximum response.

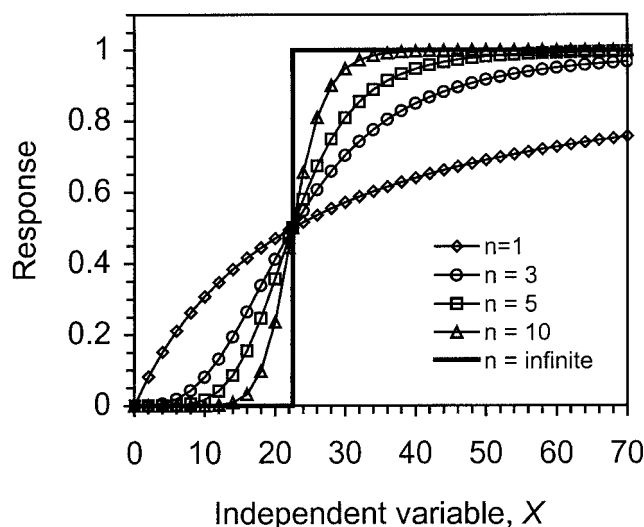


Fig. 1. Responses of the Morgan–Mercer–Flodin function (Eq. [1]) for differing values of the shape coefficient (n) with $a = 0$ (intercept), $b = 1$ (asymptote), and $X_{0.5} = 22.5$ (half of the maximum response).

Equation [1] is a general function that can take the form of a rectangular hyperbola when $n = 1$, the Hill equation (Hill, 1913) when $a = 0$, and the Michaelis–Menten equation (Michaelis and Menten, 1913) when $a = 0$ and $n = 1$.

For the vernalization response function $[f(V)]$, X is VD and Y is the vernalization response that varies from 0 to 1, with 0 corresponding to unvernallized plants and 1 corresponding to fully vernalized plants. It is assumed that fully vernalized plants do not respond to further exposure to vernalizing temperatures (Flood and Halloran, 1986). Because the response function varies from 0 to 1, the coefficients a and c in Eq. [1] have values of 0 and 1, respectively. The coefficient $VD_{0.5}$ is defined analogously to the coefficient $X_{0.5}$ as the VD when the response is one-half of the response of fully vernalized plants, i.e., when $f(V)$ is 0.5. Analysis of published data (Chujo, 1966; Wang et al., 1995; Fowler et al., 1996; Mahfoozi et al., 2001) indicates that wheat is half vernalized at about 20 to 25 VD. From these data, $VD_{0.5}$ was set to 22.5 VD. By varying the coefficient n , the MMF function can assume a variety of shapes (Fig. 1). If $n = 1$, the response curve is hyperbolic. As the coefficient n increases, the response becomes increasingly sigmoidal, increasing in steepness until it becomes a step function when n approaches infinity. Based on our understanding about the response of winter wheat to vernalization, a value of $n = 5$ was selected to describe the vernalization response to VD. When $n = 5$, the response is close to zero at values < 8 to 10 VD (at 10 VD, the response is 0.02) and > 0.98 at values > 50 VD. With these assumptions, the vernalization response function, $f(V)$, becomes:

$$f(V) = (VD)^5 / [(22.5)^5 + (VD)^5] \quad [2]$$

Equation [2] is suggested as a general function to describe the vernalization response in winter wheat.

Data Sources

Two widely accepted approaches to quantify the vernalization response in wheat are to measure the time to anthesis (calendar time or thermal time) and number of organs (FLN and/or spikelet number) in plants exposed to treatments of different VD (Chujo, 1966; Hay and Kirby, 1991; Rawson et al., 1998). The FLN approach directly reflects differences in the

Table 1. Winter wheat cultivars and source of the trials used to evaluate Eq. [2].

Cultivar	Origin	Source of data
Arapahoe	Nebraska, USA	Growth chamber experiment
Karl 92	Kansas, USA	Growth chamber experiment
Pioneer 2548	Indiana, USA	Wang et al. (1995), Table 1, several LN†
Cappelle	France	Brooking (1996), Fig. 2
Osprey	Australia	Rawson et al. (1998), Fig. 4a
Hume	South Dakota, USA	Fowler et al. (1996), Fig. 2 and Table 2
Norstar	Alberta, Canada	Fowler et al. (1996), Fig. 2 and Table 2
Augusta	New York, USA	Fowler et al. (1996), Fig. 2 and Table 2
Bezostaja	Russia	Fowler et al. (1996), Fig. 2 and Table 2
Frederick	Michigan, USA	Fowler et al. (1996), Fig. 2 and Table 2
Cheyenne	Nebraska, USA	Fowler et al. (1996), Fig. 2 and Table 2
Winalta	Alberta, Canada	Fowler et al. (1996), Fig. 2 and Table 2
Cappelle	France	Fowler et al. (1996), Fig. 2 and Table 2

† LN, leaf number of the plants at the onset of the vernalization treatment.

timing of the transition from vegetative to reproductive development (Hay and Kirby, 1991) and was used in the remainder of this paper.

To evaluate the performance of Eq. [2], independent data of FLN at different VD treatments of 12 winter wheat cultivars from 19 trials were used. The sources of these trials are presented in Table 1. These are data from published research and from a growth chamber experiment conducted as part of this study, representing wheat cultivars developed in several regions of the world and with greatly different vernalization responses.

Published research data of FLN were in tables and figures in the original papers. Data from figures were extracted by enlarging the diagram and estimating FLN values. Also, when using the published data, only experiments conducted at temperatures between 2 and 5°C were used. The daily vernalization rate [$f_{vn}(T)$, units of day^{-1}] for these experiments was calculated using a β function (Wang and Engel, 1998):

$$f_{vn}(T) = [2(T - T_{\min})^{\alpha}(T_{\text{opt}} - T_{\min})^{\alpha}(T - T_{\min})^{2\alpha}] / (T_{\text{opt}} - T_{\min})^{2\alpha}, \text{ for } T_{\min} \leq T \leq T_{\max}$$

$$f_{vn}(T) = 0 \text{ for } T < T_{\min}, \text{ for } T > T_{\max} \quad [3]$$

$$\alpha = \ln 2 / \ln[(T_{\max} - T_{\min}) / (T_{\text{opt}} - T_{\min})] \quad [4]$$

where $T_{\min}$, T_{opt} , and $T_{\max}$ are the cardinal temperatures for vernalization (minimum, optimum, and maximum) and T is the temperature at which the experiment was conducted. The cardinal temperatures for vernalization were assumed to be -1.3, 4.9, and 15.7°C, respectively (Porter and Gawith, 1999). The VD treatments were calculated by:

$$\text{VD} = \sum f_{vn}(T) \quad [5]$$

The growth chamber data set came from an experiment conducted at the University of Nebraska, Lincoln, NE, from 2 Nov. 1999 to 30 Apr. 2000. This experiment was designed to evaluate the developmental response of two winter wheat cultivars (Arapahoe and Karl 92) to VD. The VD treatments consisted of seven VD: 0, 10, 20, 30, 40, 50, and 60. The experimental design was a split-plot design with cultivars as main plots and VD as subplots. Eight pots (17 cm in diameter $\times$ 40 cm in height) were used for each cultivar-VD combination, with two plants pot^{-1} . Each pot was one replication. The photoperiod was 20 h throughout the entire growing period, and the light and dark temperatures were 25 and 25°C, respectively, before and following the vernalization period and 10 and 8°C, respectively, during the vernalization treatment. The growth chamber used in this study was not capable of operating at the optimum temperature for vernalization. Because the temperature during the vernalization treatment was

greater than the optimum, daily vernalization rate for the growth chamber experiment was calculated using the β function (Eq. [3] and [4]) with minimum, optimum, and maximum temperatures of -1.3, 4.9, and 15.7°C, respectively (Porter and Gawith, 1999), and VD was calculated with Eq. [5]. Plants were grown with adequate water and nutrients. The onset of the vernalization treatment was when plants had two fully expanded leaves. The FLN on the main stem was measured on all plants.

Data on FLN of the data sets presented in Table 1 were normalized to obtain a 0 to 1 response, representing unvernallized and fully vernalized plants, respectively, by:

$$\text{NFLN} = (\text{FLN}_{0\text{VD}} - \text{FLN}) / (\text{FLN}_{0\text{VD}} - \text{FLN}_L) \quad [6]$$

where NFLN is the normalized FLN; $\text{FLN}_{0\text{VD}}$ is the FLN of unvernallized plants, i.e., at 0 VD; FLN is for a given VD treatment; and FLN_L is the FLN at the longest VD treatment. Plants at the longest VD treatment were assumed to be fully vernalized because the FLN at this treatment had values similar to the FLN at immediately one or two shorter VD treatments.

The NFLN data were compared with the $f(V)$ predicted by Eq. [2]. For cultivars Arapahoe and Karl 92, the three-stage linear function was included in the analysis. In the three-stage linear vernalization function, the value of the coefficient VD_b is 9.2 and 8 VD, and VD_{full} is 46 and 40 VD for Arapahoe and Karl 92, respectively (Atak, 1997; Xue, 2000). For the other genotypes, there was no available data on VD_b and VD_{full} , and the three-stage linear function was not evaluated.

The root mean square error (RMSE) was calculated and used as a measure of the performance of Eq. [2] (Janssen and Heuberger, 1995):

$$\text{RMSE} = [\sum (p_i - o_i)^2 / N]^{0.5} \quad [7]$$

where p is predicted data, o is observed data, and N is the number of observations. The RMSE expresses the spread in $p_i - o_i$ and has the same units as the predicted and the observed data (in this study, it is unitless). The lower the RMSE is, the better the model.

RESULTS

The vernalization response of Arapahoe and Karl 92 obtained in the growth chamber experiment, the MMF function (Eq. [2]), and the three-stage linear function (Xue, 2000) are illustrated in Fig. 2. The MMF function described the developmental response to VD better (RMSE = 0.032) than the three-stage linear functions (RMSE = 0.060 for Karl 92 and RMSE = 0.129 for Arapahoe), which were cultivar specific. The superior performance of the MMF function compared with the three-stage linear function is particularly evident at intermediate values of VD (20–30 VD) where the linear functions underestimated the observed data for both cultivars. Also, the abrupt transitions of the linear model did not reflect the trend in the observed data, which clearly showed a smooth transition. These smooth transitions were captured by the MMF function.

The vernalization response of the 12 cultivars from the 19 trials (Table 1) along with the $f(V)$ predicted with the MMF function (Eq. [2]) are illustrated in Fig. 3. The RMSE of the estimate is 0.083, and the vernalization response is described well by the MMF function. The general trend of the observed data show a smooth transition at the beginning and end of the response, i.e., in the range of about 8 to 20 VD and after about 35 VD, and a quasi-linear response at intermediate values (20–35 VD). After 50 VD, observed data show that

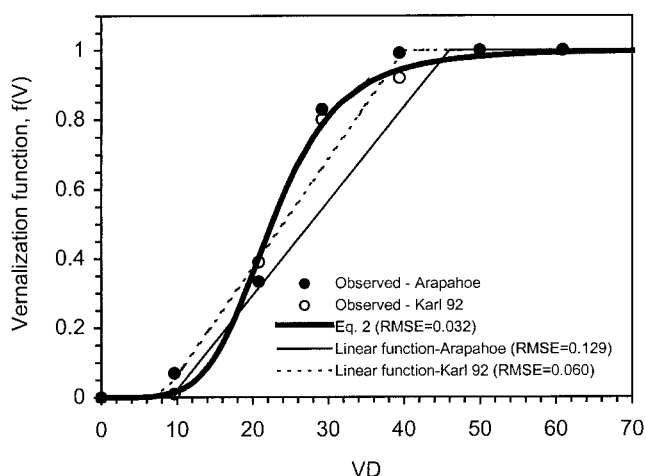


Fig. 2. The vernalization responses of the Morgan–Mercer–Flodin function (Eq. [2]) and the three-stage linear functions (linear function) used by Xue (2000), based on Wang and Engel (1998), for two winter wheat cultivars, Arapahoe and Karl 92. Observed data are from a controlled growth chamber experiment and represent the fraction of full vernalization: 0 = unvernallized plants and 1 = fully vernalized plants. VD, effective vernalization days.

plants are fully vernalized and so does the MMF function. There are three outliers in Fig. 3 at 21 VD for the cultivars Augusta, Frederick, and Winalta. While it is possible that these cultivars have a longer initial exposure requirement to vernalizing temperature than the other cultivars, an experimental error in these data cannot be ruled out as winter wheat cultivars typically have a vernalization response >0.2 at 21 VD (e.g., Wang et al., 1995; Brooking, 1996; Rawson et al., 1998; Mahfoozi et al., 2001). Excluding these three outliers, the RMSE becomes 0.065, which is similar to the RMSE of the three-stage linear function for Karl 92 and better (lower RMSE) than the three-stage linear function for Arapahoe (Fig. 2).

DISCUSSION

Several response functions used in crop simulation models depend on genotype. This creates a problem when a model needs to be used with genotypes that have unknown coefficients. Also, the use of Occam's Razor in crop modeling is encouraged (Sinclair and Muchow, 1999), i.e., the simplest theory is preferred to more complex ones or explanations of phenomena should be in terms of known quantities. Therefore, the search for generalized response functions is a major goal in crop modeling. As shown in this paper, the response of wheat development to vernalization in a wide range of genotypes can be described by a general vernalization function (Eq. [2]). Several reasons contribute to the adoption of Eq. [2] as a generalized vernalization function in winter wheat.

First, the highly predictive ability of Eq. [2] in describing a wide range of vernalization responses of different cultivars (Fig. 3) and its superior performance compared with the current genotype-specific three-stage linear function (Fig. 2) indicate its robust and general nature.

Second, this function (Eq. [2]) describes what is currently accepted in terms of vernalization response (Sla-

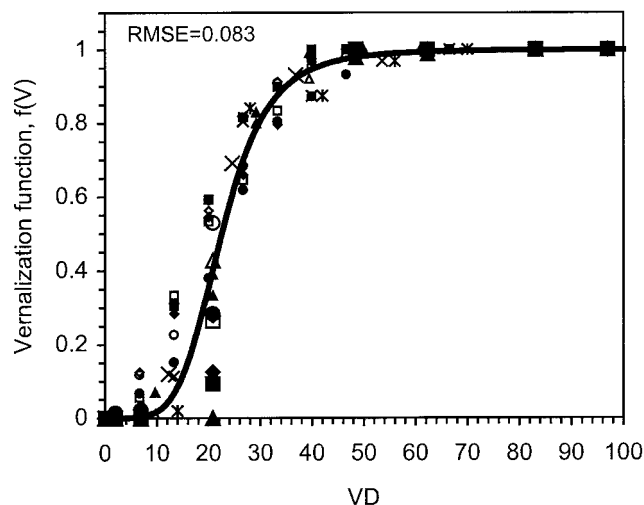


Fig. 3. The vernalization responses of 12 winter wheat cultivars from 19 trials, and the $f(V)$ predicted by the Morgan–Mercer–Flodin function (Eq. [2]). The sources of the observed data are in Table 1, and each point represents the fraction of full vernalization: 0 = unvernallized plants and 1 = fully vernalized plants. LN is the leaf number of plants at the onset of the vernalization treatment for cultivar Pioneer 2548. ● = Pioneer 2548 – LN = 1, ○ = Pioneer 2548 – LN = 2, ■ = Pioneer 2548 – LN = 3, □ = Pioneer 2548 – LN = 4, ◆ = Pioneer 2548 – LN = 5, ◇ = Pioneer 2548 – LN = 6, ▲ = Arapahoe, △ = Karl 92, × = Osprey ($T = 3^{\circ}\text{C}$), * = Osprey ($T = 5^{\circ}\text{C}$), ● = Hume, ○ = Norstar, ■ = Augusta, □ = Bezostaja, ◆ = Frederick, ◇ = Cheyenne, ▲ = Winalta, △ = Cappelle (Fowler et al., 1996), × = Cappelle (Brooking, 1996). VD, effective vernalization days.

fer and Rawson, 1994; Cao and Moss, 1997). A short period of exposure to vernalizing temperatures (<8 – 10 VD) leads to plants behaving as if they never were exposed to vernalizing temperatures. An exposure to more than 10 VD causes the plant to respond and behave differently than unvernallized plants. After 50 VD, the plant is fully vernalized, i.e., there is no further response to VD.

Third, the values of the coefficients in Eq. [2] have biological meaning. Coefficients a and c represent the developmental response of unvernallized and fully vernalized plants, respectively. The coefficient $VD_{0.5}$ represents the VD when plants show half of the response of fully vernalized plants. The shape coefficient n gives the expected sigmoidal response to VD, i.e., the response is close to zero at values <10 VD and close to one at values >50 VD.

Fourth, the response of Eq. [2] is more realistic than the three-stage linear model (Fig. 2). Biological systems are more likely to respond to environmental factors in a smooth and continuous fashion rather than in a combination of linear functions that introduce abrupt changes in the response (Shaykewich, 1995). One can argue that a three-stage linear function can also describe the data in Fig. 3. Straight lines could be fitted to the sigmoidal curve in Fig. 3. Such a linear function would also be independent of genotype and with only two coefficients (VD_b and VD_{full}). However, this approach to a generalized three-stage linear vernalization response function is not appropriate for several reasons. First, determining VD_b and VD_{full} would only be possible after the data have been analyzed. All of the coefficients used

in the MMF function (Eq. [2]) were defined independently of the experimental data presented in Fig. 3. This would not be the case with the above suggestion of using a three-stage linear function. Second, if one fits a straight line to the sigmoidal curve in Fig. 2, the resulting straight line and associated VD_b and VD_{full} would differ from the linear responses determined from an independent experiment (Atak, 1997).

Crop simulation models are incomplete tools, and errors in the predictions are still quite frequent. For example, Xue (2000) reported errors in the predictions of the date of double ridge and terminal spikelet of winter wheat cultivars of up to 19 and 10 d, respectively. The fact that the response function presented in this paper was a better predictor of the vernalization response compared with the three-stage linear function strongly suggests that predictions of developmental stages in wheat models can be improved. This hypothesis will be tested in a future paper.

CONCLUSIONS

We demonstrated that the developmental response of winter wheat to VD can be modeled by a general nonlinear function that has coefficients with biological meaning. The implication of these results is that most of the genetic variation of the vernalization response encountered among cultivars can be accounted for by using a single function, thus reducing the input data set necessary for winter wheat simulation models. This is an improvement over existing models, which use a three-stage linear function with genotype-dependent coefficients. We conclude that the MMF function as expressed in Eq. [2] can be used as a generalized vernalization function for winter wheat.

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